

Basics 25: Conservation Laws in Magnetohydrodynamics

IV - Magnetohydrodynamics

Integral balances of mass, momentum, energy, magnetic flux, and helicity

Textbook draft, version 1.0. Conservation laws are both physical principles and numerical acceptance tests. This chapter develops the local and integral balances of ideal MHD, distinguishes exact invariants from boundary-dependent ones, and shows how conservation constrains multiphysics coupling. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Convert local conservation equations into integral balances.
- Interpret momentum flux and Maxwell stress.
- Derive ideal-MHD total energy conservation.
- State the conditions for magnetic-flux and helicity conservation.
- Distinguish global conservation from local redistribution.
- Design conservation diagnostics for coupled solvers.

2 Prerequisites and reading strategy

- Basics 3, 12, 14, and 24.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Generic local and integral balance

A local conservation law is

$$\partial_t U + \nabla \cdot \mathbf{F} = S. \quad (3.1)$$

Integrating over a fixed control volume V gives

$$\frac{d}{dt} \int_V U \, dV = - \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A} + \int_V S \, dV. \quad (3.2)$$

A quantity can change inside V because it crosses the boundary or because a true source acts. Numerical diffusion should not be silently reported as a physical source.

4 Mass conservation

Ideal MHD satisfies

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (4.1)$$

For an impermeable or periodic boundary,

$$\frac{d}{dt} \int_V \rho dV = 0. \quad (4.2)$$

In a multiphysics code, particle birth, wall loss, or source modules must appear explicitly so that the same particles are not removed or deposited twice.

5 Momentum and stress transport

The momentum equation can be written

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot \mathbb{T} = 0, \quad (5.1)$$

where

$$\mathbb{T} = \rho \mathbf{u} \mathbf{u} + \left(p + \frac{B^2}{2\mu_0} \right) \mathbf{I} - \frac{\mathbf{B} \mathbf{B}}{\mu_0}. \quad (5.2)$$

The tensor contains advective momentum flux, gas pressure, magnetic pressure, and directional magnetic tension. Total momentum is conserved only when boundary tractions and external forces vanish.

6 Total energy

The conserved ideal-MHD energy density is

$$E = \frac{\rho u^2}{2} + \frac{p}{\gamma - 1} + \frac{B^2}{2\mu_0}. \quad (6.1)$$

Its flux is

$$\mathbf{F}_E = \left(E + p + \frac{B^2}{2\mu_0} \right) \mathbf{u} - \frac{\mathbf{B}(\mathbf{u} \cdot \mathbf{B})}{\mu_0}. \quad (6.2)$$

The magnetic contribution can also be interpreted through Poynting flux $\mathbf{E} \times \mathbf{B}/\mu_0$ with ideal $\mathbf{E} = -\mathbf{u} \times \mathbf{B}$. Energy may exchange among kinetic, internal, and magnetic reservoirs while total

energy remains constant.

7 Magnetic flux conservation

For a material surface $S(t)$ moving with ideal-MHD velocity,

$$\frac{d}{dt} \int_{S(t)} \mathbf{B} \cdot d\mathbf{A} = 0. \quad (7.1)$$

This is Alfvén’s flux-freezing theorem. It constrains magnetic topology under smooth ideal evolution. It does not imply that B is constant: compression and stretching alter field strength while preserving flux through the moving surface.

8 Magnetic and cross helicity

Magnetic helicity is

$$H_M = \int_V \mathbf{A} \cdot \mathbf{B} dV. \quad (8.1)$$

Its gauge-invariant interpretation requires suitable closed or relative-helicity boundary conditions. Cross helicity,

$$H_C = \int_V \mathbf{u} \cdot \mathbf{B} dV, \quad (8.2)$$

is conserved in restricted ideal barotropic settings. These invariants characterize topology and alignment, but their conservation conditions are narrower than those of mass.

9 Entropy and irreversible physics

For adiabatic ideal flow,

$$\frac{D}{Dt} \left(\frac{p}{\rho^\gamma} \right) = 0. \quad (9.1)$$

Resistivity, viscosity, shocks, heat conduction, and collisions generate entropy. A mathematically conservative shock-capturing method can conserve total energy while converting resolved kinetic or magnetic energy into internal energy.

10 Conservation across module interfaces

Suppose a fast-particle module reports lost energy $\Delta E_{\text{fast}} < 0$. A wall or thermal module should receive exactly $-\Delta E_{\text{fast}}$ minus any explicitly exported energy:

$$\Delta E_{\text{fast}} + \Delta E_{\text{wall}} + \Delta E_{\text{other}} = 0. \quad (10.1)$$

Similar ledgers apply to particle number and momentum. Conservative remapping and synchronized timestamps are therefore part of physics correctness, not merely software design.

11 Computational laboratory

Run the reduced Alfvén benchmark or a simple periodic MHD wave and track kinetic, magnetic, internal, and total energy. Perturb interpolation between two grids and measure the conservation defect. Implement a module-level particle and energy ledger.

12 Summary

- Conservation laws separate transport through boundaries from true sources.
- MHD momentum is transported by fluid and Maxwell stresses.
- Total energy can remain conserved while its physical reservoirs exchange energy.
- Flux and helicity invariants require ideal dynamics and appropriate boundaries.
- Coupled simulation modules require explicit particle, momentum, and energy ledgers.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
U	conserved density	Quantity per volume.
$\mathbf{F}$	flux	Transport rate per area.
$\mathbf{T}$	momentum-flux tensor	Fluid plus magnetic stress.
H_M	magnetic helicity	$\int \mathbf{A} \cdot \mathbf{B} dV$.
H_C	cross helicity	$\int \mathbf{u} \cdot \mathbf{B} dV$.
S	source density	Nonconservative production or removal.

13 Exercises

1. Derive the integral mass balance from the local equation.
2. Identify every term in the MHD momentum-flux tensor.

3. Rewrite the electromagnetic part of the energy flux using ideal Ohm's law.
4. State boundary conditions required for global momentum and energy conservation.
5. Design a conservation test for alpha-particle energy transferred to thermal plasma and wall loads.

14 References

References

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